

Yahya Masri

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About Me

Yahya Masri is an undergraduate in Computer Science at George Mason University with hands-on experience building AI systems and deploying end-to-end machine learning pipelines. He currently works as an AI Software Engineer Intern at Lillup, where he designs local LLM inference pipelines using llama.cpp, builds RAG systems with ChromaDB, and produces structured JSON outputs for reliable downstream integration.

As a research assistant at the NSF Spatiotemporal Innovation Center, Yahya has designed and deployed scalable data pipelines, evaluated language models on domain-specific tasks, and contributed to digital-twin frameworks for infrastructure-scale systems. He has authored multiple peer-reviewed publications on LLM benchmarking, retrieval-augmented generation, and automated data collection, published in venues including the International Journal of Digital Earth and MDPI Algorithms.

Yahya's technical strengths include Python, C, PyTorch, and experience with LLM tooling (llama.cpp, RAG architectures, prompt engineering, and model evaluation). He has built AI-driven platforms spanning marketing media generation, sports analytics, and temperature prediction, reflecting a strong emphasis on translating ideas into functional, deployed systems. He also mentors student researchers and serves as a teaching assistant for low-level programming at GMU.

Industry Experience

AI Software Engineer Intern, Lillup

2026 – Present

- Designing and implementing a local LLM inference pipeline using on-device models (via llama.cpp), with emphasis on low-latency and controllable AI behavior for practical guidance use cases.
- Building a Retrieval-Augmented Generation (RAG) system with ChromaDB to ingest, embed, and retrieve domain-specific knowledge for grounded and context-aware responses.
- Producing structured, JSON-based model outputs to enable deterministic behavior, downstream integration, and future evaluation of response quality and reliability.

Research Experience

Research Assistant, NSF Spatiotemporal Innovation Center, GMU

2024 – Present

Focus on language model evaluation and computing-infrastructure digital twins.

Advisor: Prof. Chaowei Yang

Teaching & Mentoring

Undergraduate Teaching Assistant (UTA), CS262 Intro to Low-Level Programming

2026 – Present

George Mason University

ASSIP Fellow Mentor

Summer 2025

Mentored a student researcher through the Aspiring Scientists Summer Internship Program (ASSIP), resulting in a successfully published abstract.

Research Interest

AI Evaluation: Rigorous evaluation of language models on domain-specific and real-world tasks.

AI Alignment & Safety: Pragmatic interpretability, scalable oversight, deception alignment, and safety evaluation beyond benchmark performance.

Digital Twins & Systems: Digital twins as testbeds for alignment and safety research, including infrastructure-scale data, logs, and system representations.

Publications

Benchmarking Small Language Models and Small Reasoning Language Models on System Log Severity Classification

- **Author:** Yahya Masri, Emily Ma, Zifu Wang, Joseph Rogers, and Chaowei Yang
- **Overview:** This paper introduces the first benchmark for evaluating small language models (SLMs) and small reasoning language models (SRLMs) on real-world Linux system logs for severity classification. The study compares zero-shot, few-shot, and retrieval-augmented generation (RAG) prompting strategies, revealing that while RAG substantially improves performance for several SLMs, certain reasoning models exhibit degradation under retrieval augmentation. The benchmark highlights architectural trade-offs relevant to real-time and infrastructure-scale deployment.
- **Venue:** *arXiv preprint*, 2026

Automating Data Collection to Support Conflict Analysis: Scraping the Internet for Monitoring Hourly Conflict in Sudan

- **Author:** Yahya Masri, Anusha Srirenganathan Malarvizhi, Samir Ahmed, Tayven Stover, Zifu Wang, Daniel Rothbart, Mathieu Bere, David Wong, Dieter Pfoser, and Chaowei Yang
- **Overview:** This paper introduces *Data*, an open-access, source-attributed Sudan conflict news corpus collected hourly, comprising 6,946 archived articles (as of Oct 25, 2024). The work presents a scalable web-scraping and archival pipeline that preserves full text and URLs for transparency, reproducibility, and downstream conflict analysis.
- **Venue:** *Cloud Computing and Data Science*, 2026

Optimizing Context-based Location Extraction by Tuning Open-source LLMs with RAG

- **Author:** Zifu Wang, Yahya Masri, Anusha Srirenganathan Malarvizhi, Tayven Stover, Samir Ahmed, David Wong, Yongyao Jiang, Yun Li, Mathieu Bere, Daniel Rothbart, Dieter Pfoser, David Marshall, and Chaowei Yang
- **Overview:** This paper presents the first systematic investigation of context-based location extraction using open-source large language models. By integrating retrieval-augmented generation (RAG), the study demonstrates substantial improvements in multi-incident and multi-location extraction accuracy, particularly in long-form, domain-specific news articles.
- **Venue:** *International Journal of Digital Earth*, 2025

Comparative Analysis of BERT and GPT for Classifying Crisis News with Sudan Conflict as an Example

- **Author:** Yahya Masri, Zifu Wang, Anusha Srirenganathan Malarvizhi, Samir Ahmed, Tayven Stover, David WS Wong, Yongyao Jiang, Yun Li, Qian Liu, Mathieu Bere, Daniel Rothbart, Dieter Pfoser, and Chaowei Yang
- **Overview:** This work introduces a systematic evaluation framework for crisis-event classification in conflict news, benchmarking GPT-style prompting approaches against BERT and BERT-large baselines. The study highlights trade-offs between classical fine-tuned models and instruction-based large language models in low-resource, high-noise settings.
- **Venue:** *MDPI Algorithms*, 2025

Talks & Presentations

Building a Computing Infrastructure Digital Twin, AGU25	Dec 2025
Automating Map-Making through Enhanced Geographic Information Extraction Using RAG with LLMs, AGU24	Dec 2024
Context-aware Location Extraction, NSF STC Industry Advisory Board (GMU)	Nov 2024
Conflict Incident Classification with BERT, ISSDS 2024	Jul 2024
Context-aware Location Extraction, NSF STC Industry Advisory Board (Harvard)	May 2024

Honours and Awards

Dean's List (Fall 2025), George Mason University
ADVANCE Initiative STEM Scholarship Fund, George Mason University

Dec 2025
Aug 2025

Education

B.S. Computer Science, George Mason University
A.S. Computer Science, Northern Virginia Community College

2025 – Present
2023 – 2025

Relevant Coursework

Data Structures
Formal Methods and Models
Introduction to Low-Level Programming
Linear Algebra

A+
A
A
A

Skills, Interest & Others

Programming Languages: Python, C, Java, L^AT_EX
Frameworks & Tools: PyTorch, Scikit-learn, Git
Languages: English, Arabic
Interests: Soccer, Chess, Reading